

MiniQuiz 9.3

Use Newton's method with $\mathbf{x}^{(0)} = \mathbf{0}$ to compute $\mathbf{x}^{(2)}$ for the following non-linear system:

$$\begin{aligned}4x_1^2 - 20x_1 + \frac{1}{4}x_2^2 + 8 &= 0 \\ \frac{1}{2}x_1x_2^2 + 2x_1 - 5x_2 + 8 &= 0\end{aligned}$$